CYBERHUNTER AI

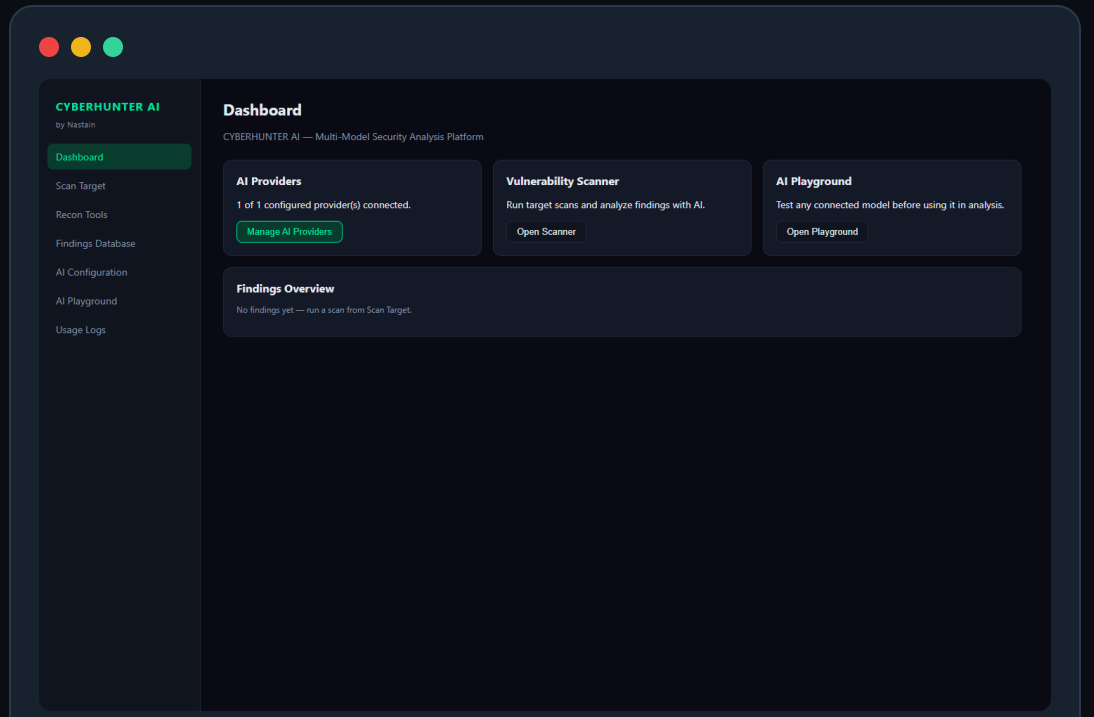
AI-Powered Automated Vulnerability Assessment Platform

Oleh **Nastain** · **Team Ternakheck**

LATAR BELAKANG

Masalah

- **Testing manual lambat** — ratusan input tersembunyi mustahil dicek satu per satu dengan tangan.
- **Scanner biasa hanya pasif** — cek header/banner saja, tanpa active probing.
- **Parameter sering terlewat** — POST body, JSON API, hidden fields, cookie, header, file .js.
- **Anti-bot menghalangi** — WAF modern memakai JavaScript challenge.
- **Hasil mentah sulit dipakai** — tanpa payload, evidence & request/response lengkap.



Dashboard — overview hasil assessment

TENTANG KARYA

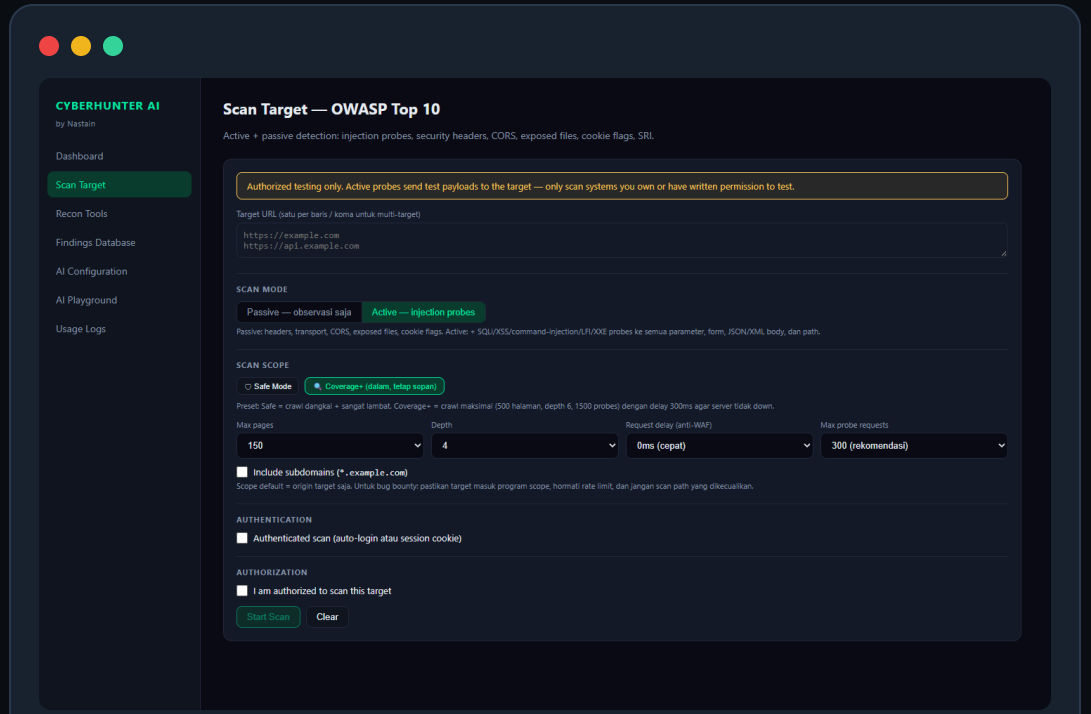
Deskripsi

- CYBERHUNTER AI adalah platform pengujian keamanan siber berbasis AI untuk mengidentifikasi, menganalisis, dan mengevaluasi kerentanan aplikasi web secara otomatis.

- Menyediakan automated vulnerability assessment pada sistem yang telah berizin diuji — konfigurasi tidak tepat, kelemahan validasi input, security headers, dan kerentanan berbasis standar OWASP.

- Hasil di dashboard interaktif: jenis kerentanan, tingkat risiko, bukti pengujian, dan rekomendasi perbaikan — dibantu AI untuk analisis dampak & prioritas mitigasi.

- **Tujuan:** efektivitas pengujian, deteksi dini, terstruktur & bertanggung jawab.



Scan console — assessment berjalan real-time

TUJUAN

Pengembangan

Efektivitas

Otomasi assessment — cakupan jauh lebih luas dari testing manual.

Deteksi Dini

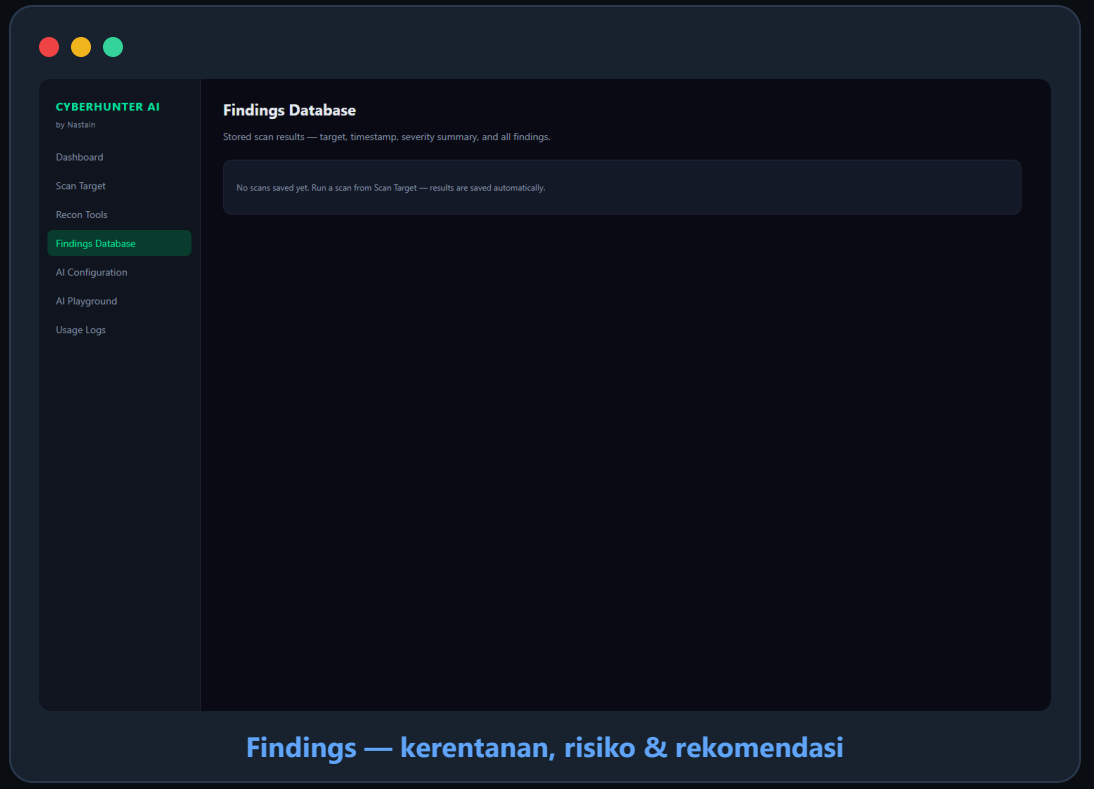
Menemukan celah sebelum dieksploitasi pihak tidak bertanggung jawab.

Terukur

Setiap temuan: payload, request/response, severity & rekomendasi.

Bertanggung Jawab

Hanya untuk sistem berizin — etika & dampak minimal.



Findings — kerentanan, risiko & rekomendasi

FITUR

Unggulan Platform

Crawler Cerdas

Multi-halaman, subdomain, form, JS bundle + parameter mining & filename mutation.

OWASP Top 10

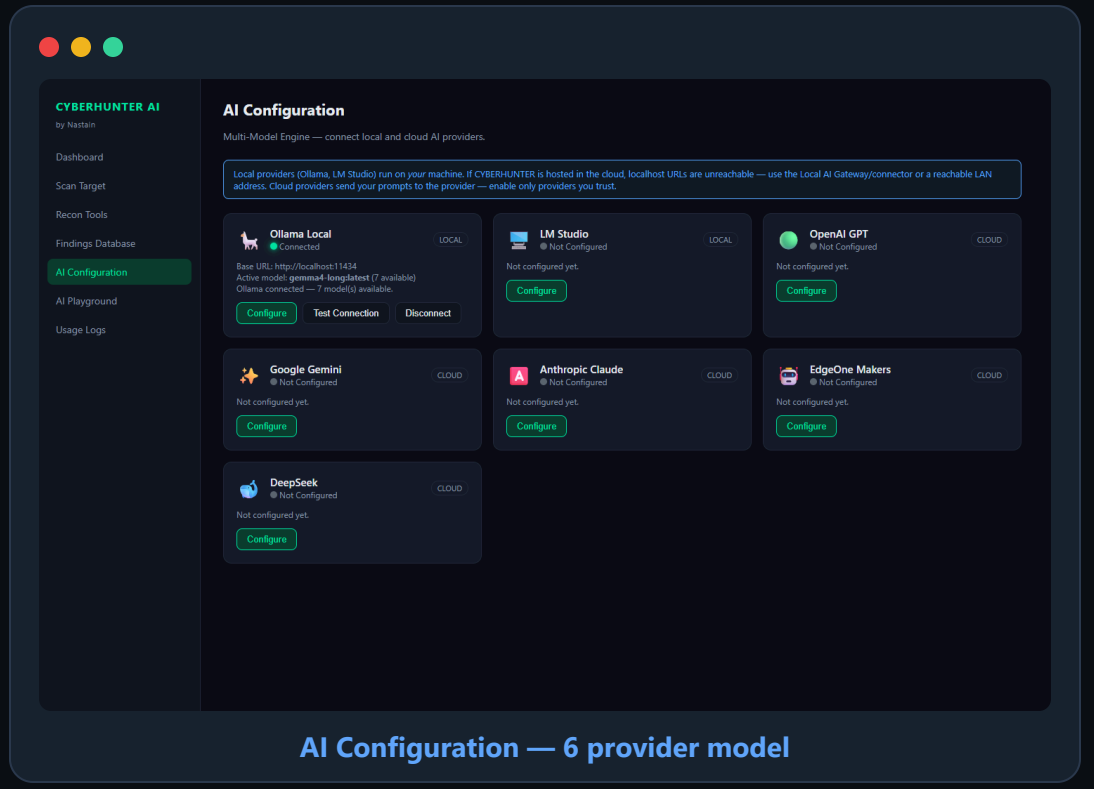
SQLi, XSS, cmdi, LFI, SSRF, XXE, CORS, headers — aktif & pasif.

Anti-Bot Solver

JS challenge (WAF splash) di-solve otomatis via headless Chromium.

AI Analysis

6 provider model — triage temuan, dampak & prioritas mitigasi.



AI Configuration — 6 provider model

CARA KERJA

Alur Assessment

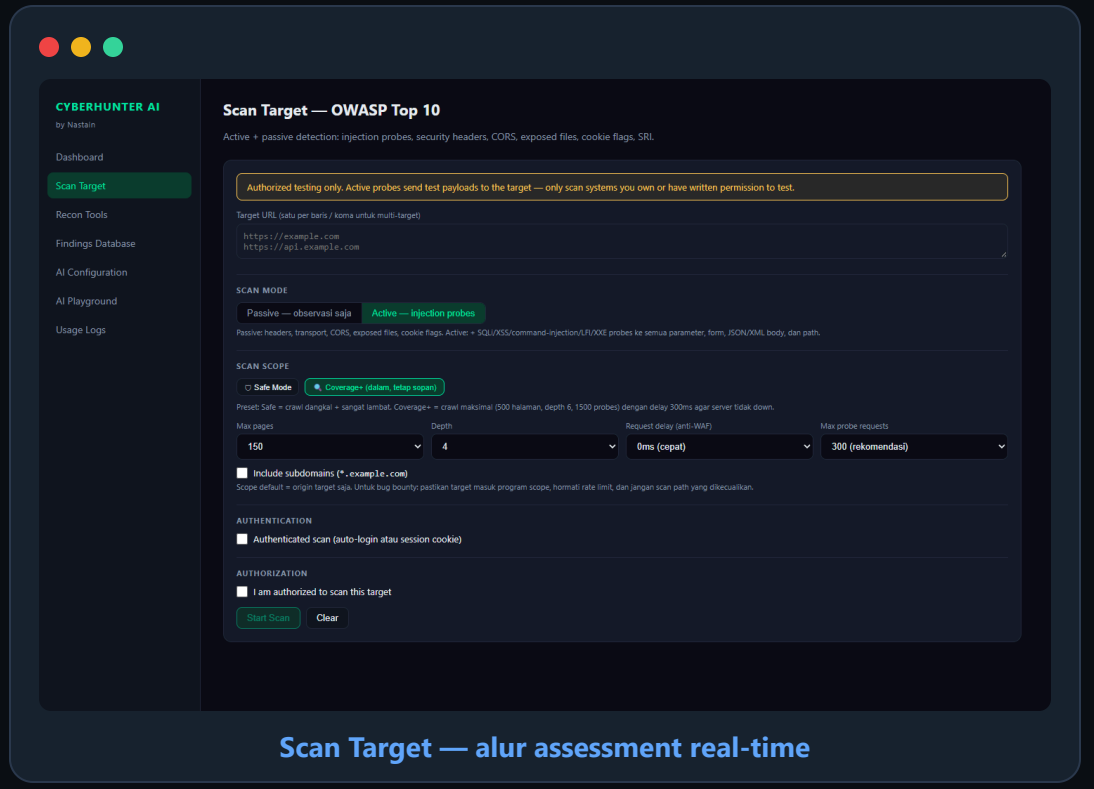
01 **Recon** Crawl halaman, form, JS, subdomain.

02 **Discover** Ekstrak semua parameter & endpoint tersembunyi.

03 **Probe** Payload OWASP aktif ke setiap input.

04 **Verify** Konfirmasi error, eksekusi JS, CVE fingerprint.

05 **Report** Evidence lengkap → dashboard + laporan bilingual.

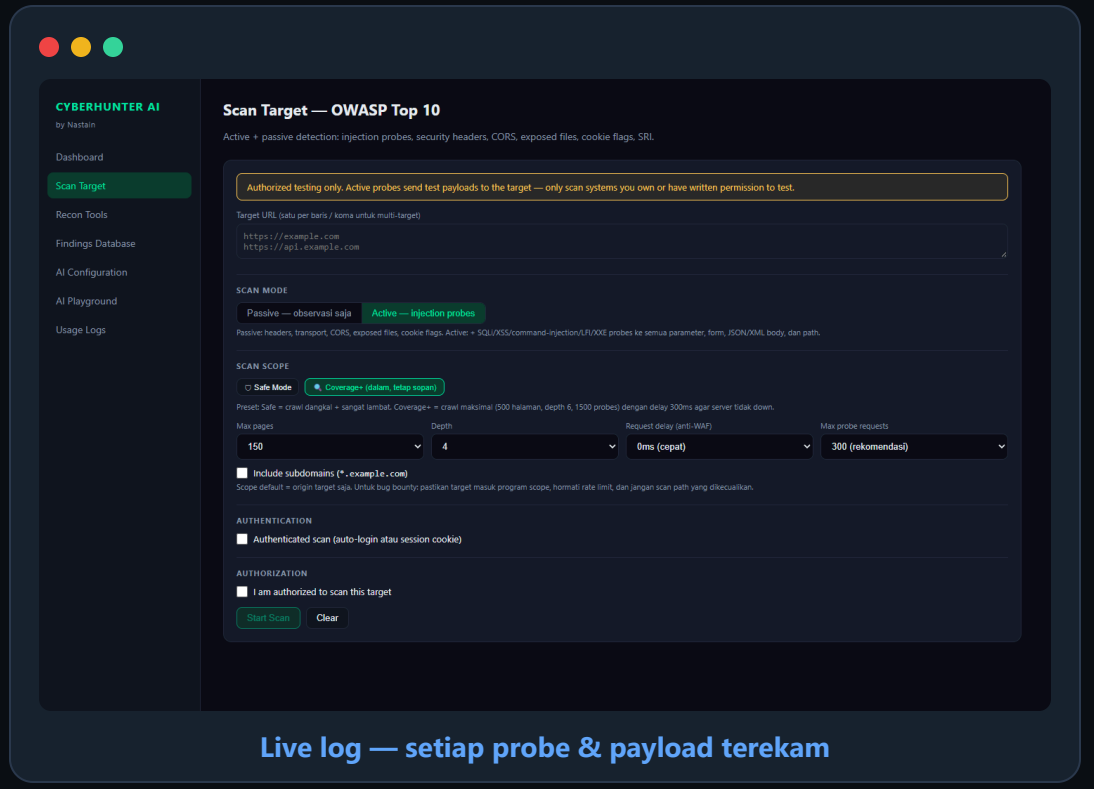


Scan Target — alur assessment real-time

COVERAGE

Semua Input Diuji

- **Input langsung** — query string, form GET/POST (input, select, textarea, hidden), JSON & XML body, headers, cookies.
- **Discovery otomatis** — parameter di file .js, ±75 nama param umum, endpoint tak ter-link, filename mutation, subdomain enum.
- **Hasil** — tidak ada input yang kelewat; semua masuk antrian probe aktif.



Live log — setiap probe & payload terekam

DETECTION

Deteksi Kerentanan

SQL Injection

Error-based, blind time-based (konfirmasi ganda), UNION extraction 15 kolom.

XSS

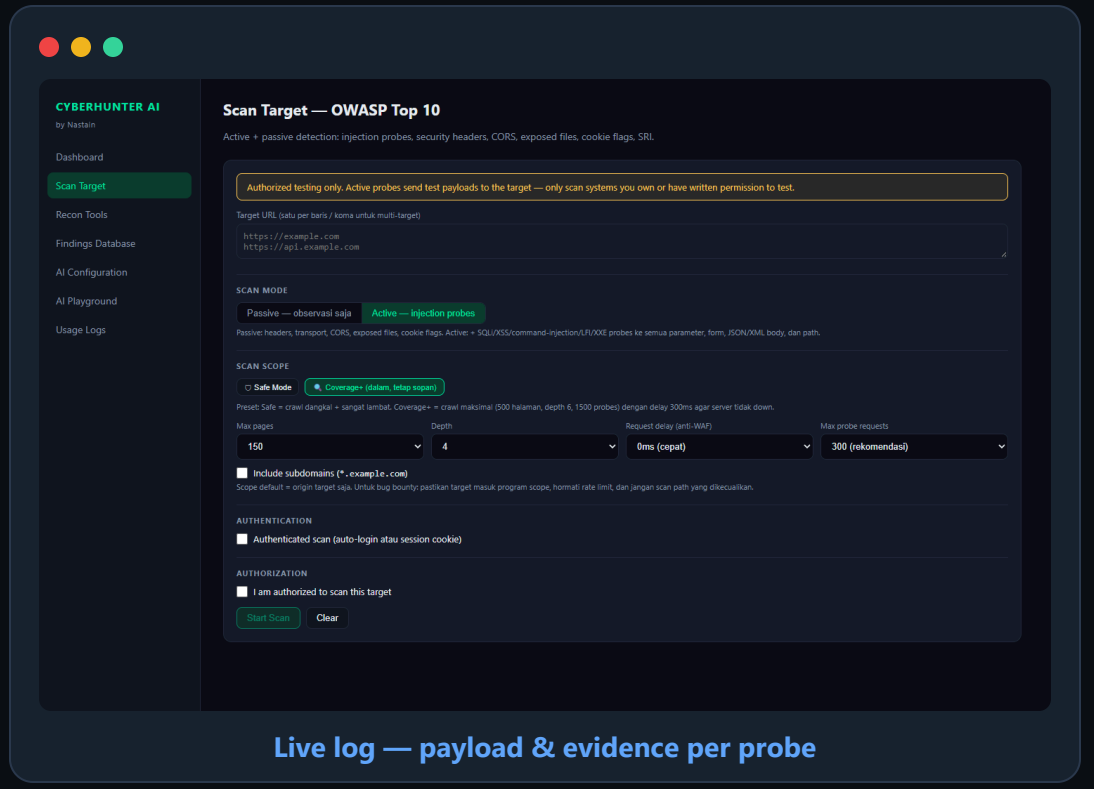
Reflected, stored & DOM — diverifikasi eksekusi JS via headless browser.

Command Injection

Marker aritmatika → output CYBERHUNTER_49, bukan refleksi biasa.

CVE Surface

Fingerprint PHP/CMS → ruleset CVE (php2shell, Drupalgeddon2, Joomla, Laravel).

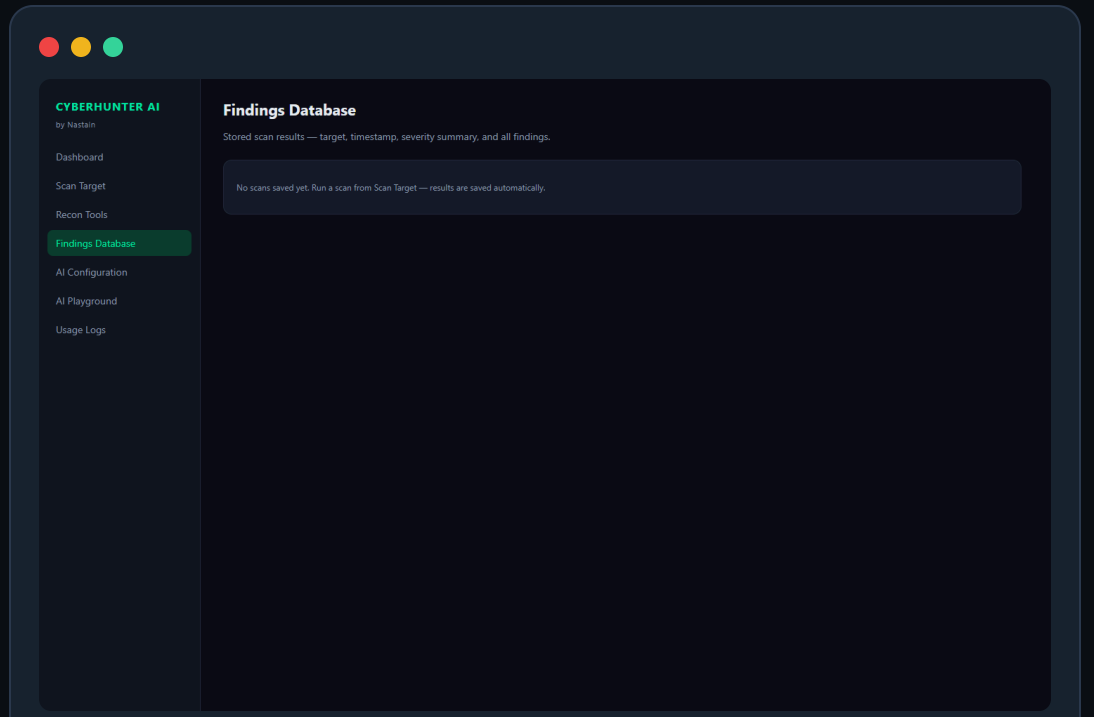


Live log — payload & evidence per probe

CVE

PHP2Shell — CVE-2026-109618 & 109619

- **Apa itu** — memory corruption & use-after-free di PHP core. Unauthenticated, ~1 request untuk mod_php.
- **Kenapa kritis** — flaw di runtime PHP; aplikasi aman pun rentan. PHP ≤8.1 (EOL) tidak akan dapat patch.
- **Deteksi** — version fingerprinting (header, PHPSESSID, phpinfo) + behavioral probe multipart.
- **Hasil** — versi rentan → CRITICAL; indikasi behavioral → HIGH. Fixed: 8.2.35 / 8.3.36 / 8.4.27 / 8.5.12.



Finding — severity & evidence lengkap

HASIL

Dashboard & Pelaporan

Jenis Kerentanan

SQLi, XSS, cmdi, LFI, SSRF, XXE, CORS, headers, CVE — terklasifikasi.

Tingkat Risiko

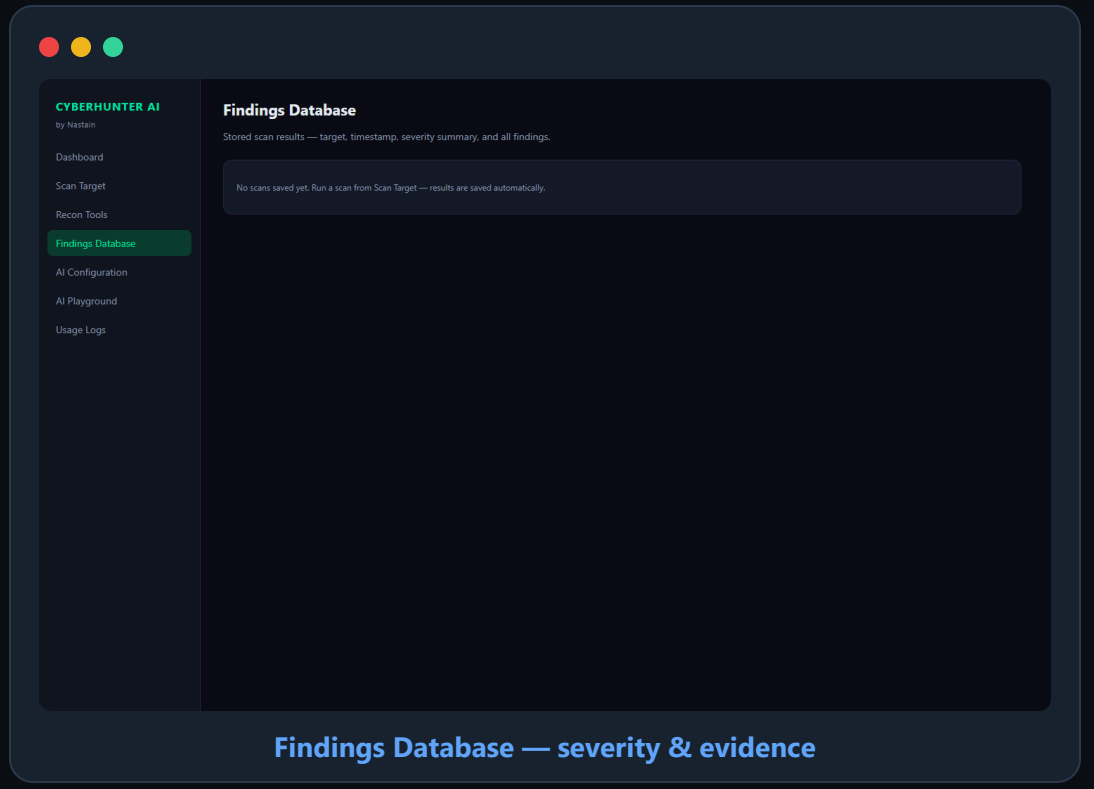
Critical/High/Medium/Low/Info — chart interaktif + filter.

Bukti

Payload persis, lokasi parameter, request & response mentah.

Rekomendasi

Panduan perbaikan + laporan HTML bilingual (EN/ID) format XRay.

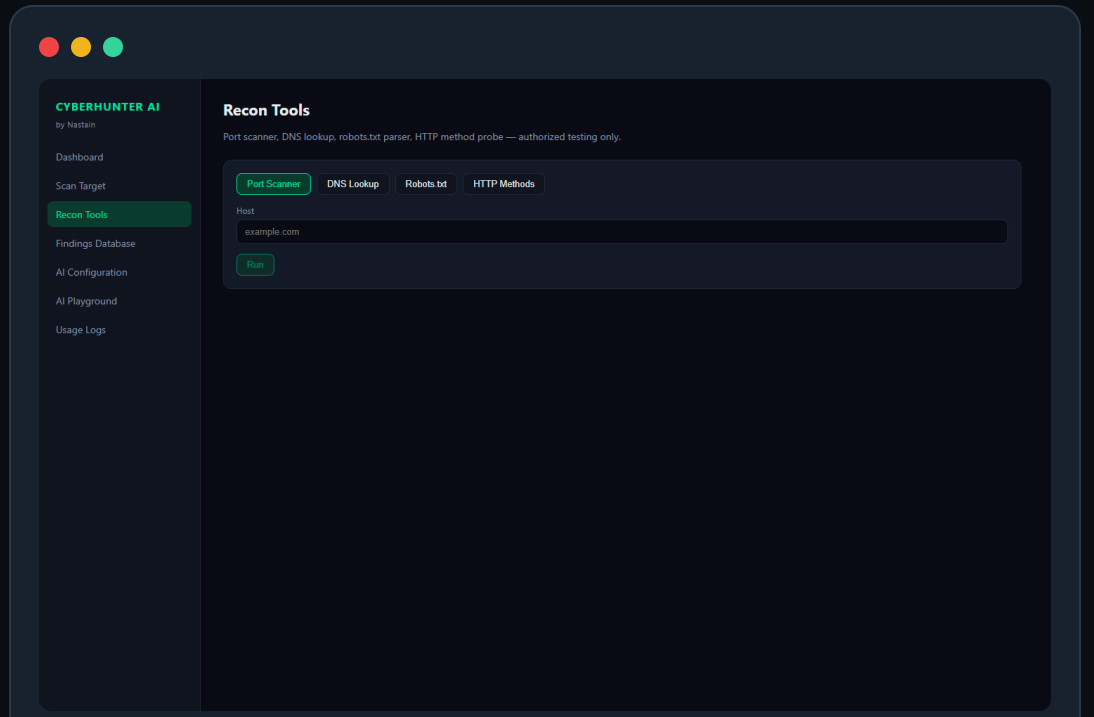


Findings Database — severity & evidence

KEAMANAN

Etika & Authorized Testing

- **Otorisasi wajib** — active probing hanya untuk sistem milik sendiri / berizin tertulis.
- **Cookie aman** — session cookie di-scope per-origin; nilai disensor di log, DB, laporan.
- **Payload non-destruktif** — marker aritmatika, canary, error-based — tanpa exploit merusak.
- **Rate limiting** — request delay dapat diatur; beban ke target tetap wajar.
- **Scope awareness** — robots.txt Disallow → peringatan; area terlarang tidak disentuh.



Tools — utilitas pendukung assessment

STACK

Teknologi

Frontend

React 18 · TypeScript · Vite — dashboard, live log, severity chart.

Backend

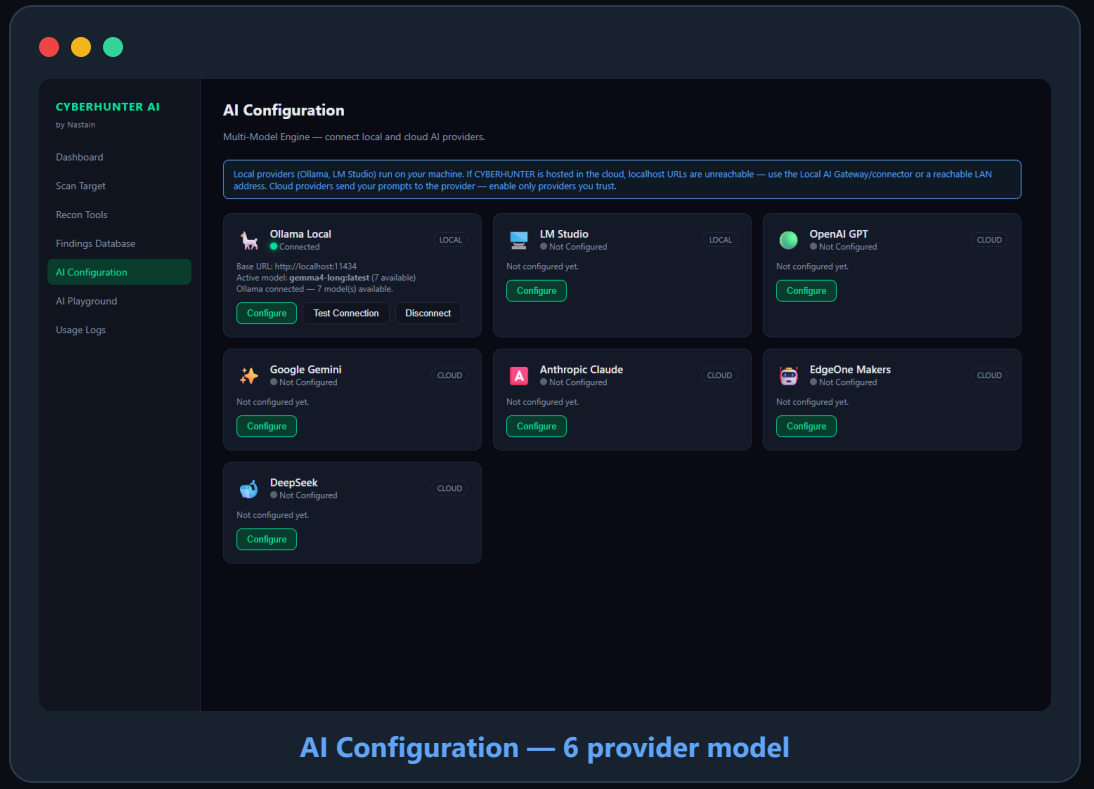
Node.js · Express — crawler, probe engine, Puppeteer.

Deteksi

±90 payload OWASP · 75+ param mining · DOM XSS verified.

AI

6 provider (cloud & lokal) — key terenkripsi, usage logs.



AI Configuration — 6 provider model

ARSITEKTUR

System Design

FRONTEND React 18 + TypeScript + Vite

Dashboard · Scan Console · Findings · AI Configuration · Laporan HTML

▼

BACKEND Node.js + Express — REST API + SSE live log

crawler.js · scanner.js · cve.js · domxss.js · challenge.js · db.js

▼

DATA & AI Findings DB (JSON persisten) + 6 AI Provider

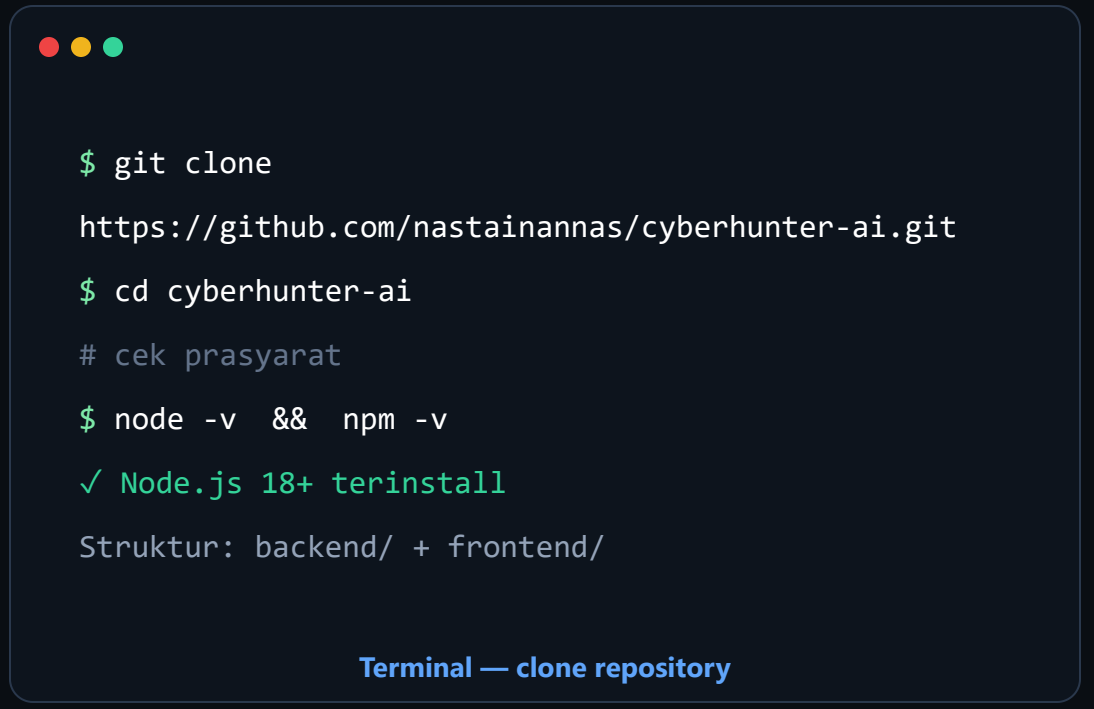
OpenAI-compatible · Anthropic · Gemini · Ollama · LM Studio · EdgeOne

Frontend → REST API / SSE → Scanner Engine → Findings DB → Report Engine → AI Analysis

INSTALASI

Cara Download

- **Repository GitHub** — github.com/nastainannas/cyberhunter-ai — public, langsung bisa di-clone.
- **Opsi 1 — Git clone** — git clone https://github.com/nastainannas/cyberhunter-ai.git lalu cd cyberhunter-ai.
- **Opsi 2 — Download ZIP** — GitHub → Code → Download ZIP → extract ke folder project.
- **Prasyarat** — Node.js 18+ dan npm — cek dengan node -v && npm -v.



Terminal — clone repository

INSTALASI

Cara Instal & Menjalankan

- **Backend dulu** — cd backend → npm install → npm start. API di http://localhost:8787
- **Frontend** — cd frontend → npm install → npm run dev. UI di http://localhost:5173
- **Pertama kali** — npm install mengunduh semua dependency (±1–2 menit).
- **Konfigurasi AI (opsional)** — AI Configuration → pilih provider → isi API key → Test Connection.
- **Mulai scan** — Scan Target → masukkan URL → mode Passive/Active → Start Scan.



Terminal — instalasi & menjalankan

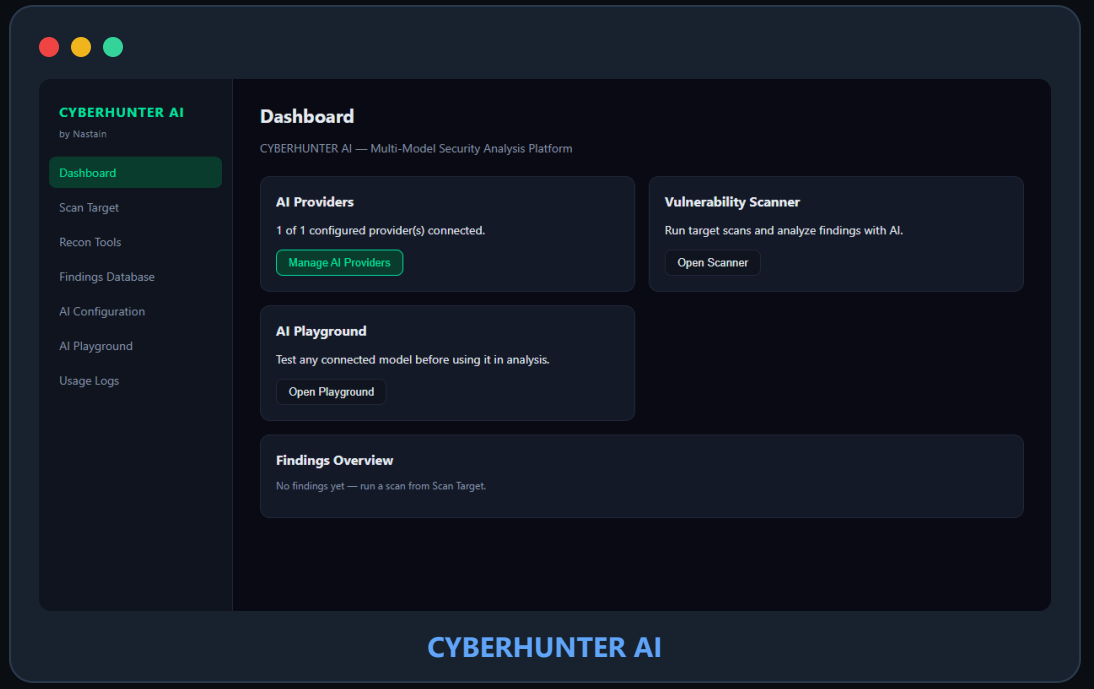
PENGEMBANGAN

Roadmap

01 **Jangka Pendek** — Template Nuclei · wizard authenticated scan · export PDF.

02 **Menengah** — Distributed queue · authenticated crawling · GraphQL fuzzing.

03 **Jangka Panjang** — Plugin custom · sinkronisasi scope bug bounty · AI triage otomatis.



CYBERHUNTER AI

Terima Kasih

CYBERHUNTER AI — AI-Powered Automated Vulnerability Assessment Platform

Scan lebih dalam, temukan lebih banyak, amankan lebih awal.

github.com/nastainannas/cyberhunter-ai

Oleh Nastain · Team Ternakheck